

AccessMatrix Google Cloud HSM Implementation Guide - 6.0.1-GA (AI)

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1. Preface - AccessMatrix Google Cloud HSM Implementation Guide

Google Cloud Platform (GCP) provides [Cloud Key Management Service \(KMS\)](#), which is a cloud-hosted service that lets you manage cryptographic keys and perform cryptographic operations securely. It provides two options on how to store and protect keys, and perform cryptographic operations, i.e. by either:

- Software
- Hardware Security Module (HSM), which allow you to host encryption keys and perform cryptographic operations securely in FIPS 140-2 Level 3 certified HSMs

[Cloud KMS](#) provides a [front end](#) to access Cloud HSM.

Intended Reader

The readers must possess some familiarity with GCP concepts like the concept of Google service accounts and others, and operations that include creating common Google cloud objects like Linux Virtual Machine (VM) and common object hierarchy like project and location.

Document Scope

While this document provides necessary steps to provision KMS keys, it will not provide documentation on operating the GCP that already has its own vast and complete documentation. However, where it is needed, this document will provide:

- Links to Google documentation to provision/manage supporting resources (e.g. VMs) that must be prepared for this integration.
- Gcloud command-line instruction for resources that are directly used by AccessMatrix (e.g. cryptographic keys). While you can manage and configure KMS keys by using various ways (e.g. GCP Console, command line), in this documentation, the instructions are provided by using the `gcloud` command.

For feedback or questions regarding this document, contact doc@i-sprint.com.

Related Documents

For more information, see the following documents in the AccessMatrix Product Suite:

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- [*AccessMatrix Common Administration Guide*](#)
 - [*AccessMatrix Common Deployment Guide*](#)
 - [*UAS Pseudo E2EEA Implementation Guide*](#)

2. Introduction to Google Cloud HSM Implementation with AccessMatrix

This document shows you how to implement and integrate AccessMatrix with Google KMS and its Cloud HSM.

Terminology

This table lists some common terms that are used in this document. It is not intended as a comprehensive definition for each item.

Term	Description
Google Cloud Platform (GCP)	Google offers its cloud platform, a serverless computing environment that provides infrastructure as a service, platform as a service including computing, data storage, data analytics and more. Its registration requires a credit card or bank account details.
Key Management Service (KMS)	Google cloud-hosted key management service that lets you manage cryptographic keys and perform cryptographic operations.
Hardware Security Module (HSM)	HSM is one of the options in KMS to host cryptographic keys and perform its operations.
Gcloud	Command-line interface installed on the workstation that allows you to do various tasks on GCP, such as creating and managing Compute Engine VMs, creating HSM / KMS resources, etc.
Object hierarchy	Hierarchical structure designed for useful and elegant access control management. Cloud KMS stores cryptographic keys under an object hierarchy. Two highest elements of object hierarchy are project and location.
Project	The highest structure in the object hierarchy. Cloud KMS resources belong to a project. You can think of a project as the organizing entity for your applications. A project is made up of the settings, permissions, and other metadata that describe your applications.
Location	It represents the geographical data centre locations, for example, asia-southeast1 (Singapore), asia-east1 (Taiwan), asia-east2 (Hong Kong), asia-northeast1 (Tokyo), asia-northeast2 (Osaka), and so on. Within a project, Cloud KMS resources can be created in multiple locations. Refer to the list of the available region on Google documentation as it might be updated from time to time. There is a unique location for KMS resources called "global". When created in the global location, your KMS resources are available from locations spread around the world.

Term	Description
Service Account	A particular type of Google account intended to represent a non-human (non-interactive) user that needs to authenticate and be authorized to access data in Google APIs.
Key Provider	It is a key manager in AccessMatrix to define communication interface with Google Cloud HSM for different implementations of cryptography key provisioning.

AccessMatrix Integration with Google Cloud HSM

Google Cloud HSM supports both symmetric and asymmetric cryptographic keys and operations. AccessMatrix system uses Cloud HSM as a key provider, which means it uses Cloud HSM to store symmetric keys and perform cryptographic operations (encryption/decryption) inside it. Cloud HSM supports AES-256 symmetric keys with Galois Counter Mode (GCM).

AccessMatrix system does not provide key management to Cloud HSM, e.g. creating the keys, destroying the keys, rotating the keys, etc. All of those key management/operations are to be done directly on Cloud KMS (either using GCP Console or command line).

Authentication to Access Google Cloud HSM

Any entity or principal that accesses any Google cloud resources must be identified and authenticated. AccessMatrix uses Application Default Credentials (ADC) strategy as the server to server authentication to access Cloud HSM.

- **Identification**

AccessMatrix identifies itself by using a [service account](#). A service account is a special kind of account that belongs to an application or a virtual machine (VM) instance, not a person.

- **Authentication**

While [Google provides several authentication strategies](#), AccessMatrix uses [Application Default Credentials \(ADC\) strategy](#) to authenticate itself to access Cloud HSM.

Out of Integration Scope

While they might be mentioned in other parts of this document, these items are not part of this integration scope, i.e.

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- Key management to Cloud HSM, e.g. creating the keys, destroying the keys, rotating the keys, etc.
 - Asymmetric keys and operations are out of scope in this integration document.
 - AccessMatrix must be deployed in Linux VM on Google Compute Engine. Thus, on-premise deployment of AccessMatrix that integrate with Cloud HSM is not supported.

Implementation Checklist

See below a list of items that must be prepared before integrating AccessMatrix with Cloud HSM.

Item	Description
GCP account	You have a valid GCP account.
Project	You have a created a GCP project.
Billing Enabled	The project must have billing enabled. Refer to Google documentation on how to enable billing for a project .
KMS Enabled	You have enabled KMS for the project. It will affect billing. Refer to Google KMS Pricing documentation for more details .
Service Account	You have created a service account, that will be used to identify AccessMatrix to access cloud HSM. Refer to Google documentation on how to create a service account . Alternatively, you can let GCP to create a service account automatically while creating a Linux VM.
Linux VM	You have a running Linux VM (to host AccessMatrix) on GCP Compute Engine with proper hardware sizing (e.g. 4 CPU core, 16 GB RAM) based on your computing needs or refer to AccessMatrix Performance Benchmark document. The Linux VM must belong to the service account (either you create it as part of Google IAM principals or automatically created for you when creating the Linux VM). To create and start a VM instance, refer to Google documentation on how to create and start a VM instance . The VM Access scopes must be set to "Allow full access to all Cloud APIs".
Gcloud Enabled	You have installed gcloud command-line tool on your workstation, a command-line interface to interact with GCP. Refer to Google documentation on how to download and install gcloud .
AccessMatrix	You have set up the above items on the Linux VM, and on Google Cloud HSM. Next, refers to AccessMatrix Google Key Provider Deployment for other configurations. The ports needed for accessing AccessMatrix must also be opened, for example, by default, TCP/8080 (http) and TCP/8443 (https).

Implementation Planning

Security Consideration

As described in Performance Consideration section below, AccessMatrix can use Cloud HSM key as wrapping key, to unwrap keys that are used by AccessMatrix to secure attributes (AEK) or PIN (PEK). Which means, during operation, the AEK or PEK will be in the clear format in AccessMatrix memory.

To be more secure, you can consider using HSM keys directly, which, on the other hand, will incur some performance penalty and cost.

Firewall Port Opening

By default, the ports of AccessMatrix VM (Linux VM where AccessMatrix is installed) in Google Virtual Private Cloud are closed, so you need to open them to access AccessMatrix. For example, the default ports of AccessMatrix are TCP/8080 (http) and TCP/8443 (https); thus, you need to open either of them or both, depend on your need.

Integration Consideration

As of version 5.6.6-GA, AccessMatrix must be deployed in Linux VM on GCP Compute Engine. On-premise deployment of AccessMatrix that integrate with Cloud HSM is not supported.

Performance Consideration

For optimum performance, ensure that the Linux VM and keys that you use are located on the same location, e.g. asia-southeast1.

AccessMatrix can use the Cloud HSM keys directly, for example as the attribute encryption key (AEK), where AccessMatrix calls KMS API to encrypt an attribute directly. Thus, every time AccessMatrix needs to have the attribute in clear text, it will call KMS API to decrypt.

Another option is to use Cloud HSM keys as wrapping keys. In this case, you can create an AEK in AccessMatrix default System Key Provider. This AEK key then will be wrapped by Cloud HSM key. Thus, AccessMatrix needs to call KMS API once to unwrap AEK. Then, this AEK can be used to encrypt/decrypt the attributes.

Depending on the use case, for example, if you use it for PseudoE2EEA login module, for faster performance, you might want to consider using Cloud HSM keys as wrapping key instead of PIN Encryption Key (PEK) directly.

3. GCP Cloud Key Management Service Configuration

Prior to the GCP Cloud KSM configuration, ensure the following prerequisite checklist has been prepared:

- A GCP project with billing and KMS enabled.
- A running Linux VM on GCP Compute Engine.
- A service account (or use the Linux VM service account).
- Gcloud, command-line tool, has been appropriately installed on your workstation.

Create Key Ring and Crypto Key

AccessMatrix uses Google Cloud HSM as a key provider. As it does not do key management by itself, you need to create symmetric keys (and the key ring) by using GCP KMS interface, either by using the GCP Console or gcloud command-line interface.

Cryptographic keys are managed in [object hierarchy](#). All resources must belong to a project. Under a project, there will be "locations". Within a project, KMS resources can be created in multiple locations.

Under a location, you can create "keyring". A key ring is a grouping of keys for organizational purposes. Keys inherit permissions from the key ring that contains them. Grouping keys with related permissions together in a key ring allows you to grant, revoke, or modify permissions to those keys at the key ring level, without needing to act on each one individually.

Keys are created inside the key ring.

All the Gcloud commands below are to be executed on the command-line interface (e.g. Linux Shell or Windows CMD) on your workstation. The sample will be using:

- location: asia-southeast1 (Singapore)
- service account: my-service-account@my-project.iam.gserviceaccount.com
- key ring: AM_KR
- key: AM_SYM_KEY

Therefore, you need to replace with your identifier accordingly.

Create a key ring, name it AM_KR, in asia-southeast1 location:

```
{gcloud kms keyrings create AM_KR --location asia-southeast1}
```

You can confirm the key ring creation by listing them, which in this case, key ring AM_KR should be listed there:

```
{gcloud kms keyrings list --location asia-southeast1}
```

When the key ring is ready, you can create the key. You don't need to specify algorithm or key length since all symmetric keys are 256-bit Advanced Encryption Standard (AES-256), with Galois Counter Mode (GCM).

```
{gcloud kms keys create AM_SYM_KEY --location asia-southeast1 \
--keyring AM_KR --purpose encryption --protection-level hsm}
```

You can confirm the key creation by listing them, which in this case, key AM_SYM_KEY should be listed there:

```
{gcloud kms keys list --location asia-southeast1 --keyring AM_KR}
```

Granting Access to Service Account

The service account used by AccessMatrix must be granted with privileges to access the newly created key. You can check the service account used by the Linux VM (that host AccessMatrix) on GCP Console, or by running this command:

```
{
gcloud iam service-accounts list
}
```

In this example, the service account is: my-service-account@my-project.iam.gserviceaccount.com. Then you can grant permissions by assigning role cloudkms.cryptoKeyEncrypterDecrypter to that account:

```
{gcloud kms keys add-iam-policy-binding AM_SYM_KEY \
--location asia-southeast1 --keyring AM_KR \
--member serviceAccount:my-service-account@my-project.iam.gserviceaccount.com \
--role roles/cloudkms.cryptoKeyEncrypterDecrypter}
```

If you can run it successfully, then you should see this output.

```
{Updated IAM policy for key [AM_SYM_KEY].
bindings:
-members:
  -serviceAccount:my-service-account@my-project.iam.gserviceaccount.com
  role: roles/cloudkms.cryptoKeyEncrypterDecrypter
etag: BwWY8KD7QKM=
version:1}
```



Note: Refer to [Google Cloud documentation](#) to learn the types of role (i.e. roles/viewer) you require to assign to the service account. Otherwise, you may encounter Permission Denied Exception.

Verify Key Readiness

Before you start any configuration on the AccessMatrix side, you must verify that the key is ready by running a simple encryption/decryption on gcloud.

Create a text file from your command line, and name it **clear.txt**. For example:

```
{echo "AccessMatrix and Google Cloud HSM"> clear.txt}
```

Encrypt it by using Cloud HSM, and ensure that **clear.txt.encrypted** is created successfully. You can open it by using any text editor to confirm that the content is ciphertext.

```
{gcloud kms encrypt --location asia-southeast1 --keyring AM_KR \  
--key AM_SYM_KEY --plaintext-file clear.txt --ciphertext-file clear.txt.encrypted}
```

Decrypt file **clear.txt.encrypted** back to clear text, name the file **clear2.txt**.

```
{gcloud kms decrypt --location asia-southeast1 --keyring AM_KR \  
--key AM_SYM_KEY --plaintext-file clear2.txt --ciphertext-file clear.txt.encrypted}
```

Check the content of **clear2.txt**, and if it is the same as **clear.txt** (the original clear text), then Cloud KMS resources configuration is accomplished.

4. AccessMatrix Google Key Provider Deployment

This part assumes AccessMatrix has been deployed properly on GCP Compute Engine (the Linux VM), and necessary ports have been opened, for example, TCP/8080 (http) and/or TCP/8443 (https). You can confirm it by opening AdminConsole, visit `http://<hostname:port>/am5` (or `https://<hostname:port>/am5`).

Deploy Google Key Provider Plugin

Google Key Provider is implemented as a plugin. Thus, you need to deploy the plugin into the AccessMatrix folder by following these steps:

1. Copy **am-googlekprov-core.jar** file from `<ACMATRIX_ROOT>\sdk\googlekprov\` folder to `<ACMATRIX_ROOT>\tomcat\webapps\am5\WEB-INF\lib` folder.
2. Copy `<ACMATRIX_ROOT>\sdk\googlekprov\lib\*.jar` files to the `<ACMATRIX_ROOT>\tomcat\webapps\am5\WEB-INF\lib` folder.
3. Copy **am-googlekprov-plugin.jar** file from `<ACMATRIX_ROOT>\sdk\googlekprov\` folder to `<ACMATRIX_ROOT>\tomcat\plugins` folder.
4. Copy **field-definitions-googleKeyProvider.xml** from `<ACMATRIX_ROOT>\sdk\googlekprov\res\` folder to `<ACMATRIX_ROOT>\tomcat\webapps\am5\WEB-INF\classes\` folder.
5. Copy **wpecustom-googleKeyProvider.properties** from `<ACMATRIX_ROOT>\sdk\googlekprov\res\` folder to `<ACMATRIX_ROOT>\tomcat\webapps\am5\WEB-INF\classes\` folder.
6. Rename **wpecustom-googleKeyProvider.properties** in the folder `<ACMATRIX_ROOT>\tomcat\webapps\am5\WEB-INF\classes\` to `wpecustom.properties`.
7. Add the following line to the file **amsystem.properties** in the folder `<ACMATRIX_ROOT>\tomcat\webapps\am5\WEB-INF\classes\`:

Properties (.properties files)

```
am.field_definition.file.0=field-definitions-  
googleKeyProvider.xml
```

8. Start AccessMatrix server, check **am.log** file (`<hostname>_am.log`) to see if Google Key Provider plugin is loaded. Find this keyword "Plugin loaded: GoogleKeyProviderPlugin". For example:

```
2019-12-03 14:51:52.429 [main] INFO - (GoogleKeyProviderPlugin.java:28) Plugin loaded:  
GoogleKeyProviderPlugin - 0.1
```

5. AccessMatrix Google Key Provider and Encryption Key Configuration

This part assumes the AccessMatrix Google Key Provider plugin has been deployed/installed successfully. You can confirm it by navigating to **Administration Dashboard > Security Policy > Key Management > Key Manager > Key Provider**, clicking the **Create** button, then checking if "Google Key Provider" is listed in the **Implementation** drop-down list.

To configure the key provider and encryption keys in AccessMatrix to use Google Cloud HSM, you should also have completed the steps detailed in the [GCP Cloud Key Management Service Configuration](#) section and generated a crypto key, as well as any necessary information of your GCP project.

Setup Steps

This section shows you how to configure the AccessMatrix system to integrate with Google Cloud HSM in the Admin Console. Before you start, it is assumed that you have completed all the prerequisites.

- **Configure Google Key Provider**

Create a key provider of Google Key Provider implementation. Refer to [AccessMatrix Common Administration Guide - Configure Key Provider](#) for details of Google Key Provider. Use this provider to integrate with Google Cloud HSM. Refer to [Configure Google Key Provider](#) for more details.

- **Configure Encryption Key**

AccessMatrix Google Key Provider supports symmetric keys for the following purposes:

- Attribute encryption
- Key derivation
- Key wrapping
- Pin encryption
- Token encryption

Create encryption keys that will be used for various purposes based on the Google Key Provider created. For example, you create encryption key with "Pin Encryption" purpose to secure the password, or you create encryption key with "Key Wrapping" purpose to encrypt the local encryption keys (that are defined in Default System Key Provider). Refer to [Configure Encryption Key](#) for more details.



Notes:

- Refer to [AccessMatrix Common Administration Guide - Configure Encryption Key](#), for an explanation on each key's purpose.

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- If the use case is for PseudoE2EEA, refer to the [UAS Pseudo E2EEA Implementation Guide](#).

Configure Google Key Provider

Upon successful deploying AccessMatrix Google Key Provider plugin, the "Google Key Provider" implementation is available in the AccessMatrix Admin Console.

1. Navigate to **Administration Dashboard > Security Policy > Key Management > Key Manager > Key Provider** on the left pane.
2. Click **Create** and select "Google Key Provider" in the **Implementation** drop-down list.
3. Click **Next**.
4. Enter a **Key Provider Id** (e.g. "GKey") and a **Key Provider Name** (e.g. "GKey") and enter its description.
5. Specify the **Key Ring Name** (e.g. "ISPRINT_KR").
6. Specify the **Location Id** (e.g. "asia-southeast1").
7. Specify the **Project Id** (e.g. "hallowed-web-259403").
8. Click **Save**. A new key provider is created.

Configure Encryption Key

1. Navigate to **Administration Dashboard > Security Policy > Key Management > Encryption Key** on the left pane.
 2. Click **Create** and select the newly created key provider (e.g. "GKey") from the **Key Provider** drop-down list.
Notice there is only one available **Encryption Key Template**, "Symmetric key (secret key)", as the AccessMatrix system integrates with Google cloud HSM to perform symmetric operations only.
 3. Click **Next**.
 4. Enter an **Encryption Key Id** (e.g. "GEncKey") and an **Encryption Key Name** (e.g. "Google Encryption Key") and enter its description.
 5. AccessMatrix system will automatically fetch the Crypto Key Name(s) you created in Google Cloud KMS. If there is more than one crypto key name in the drop-down list box, select a crypto key name.
 6. Next, select key purpose from the ***Purpose** drop-down list on the **Attributes** tab.
-

7. Click **Save**. An encryption key is created.



Note: You can click the **Test KeyInfo** and **View Public Key** buttons to check if this key is created successfully.
